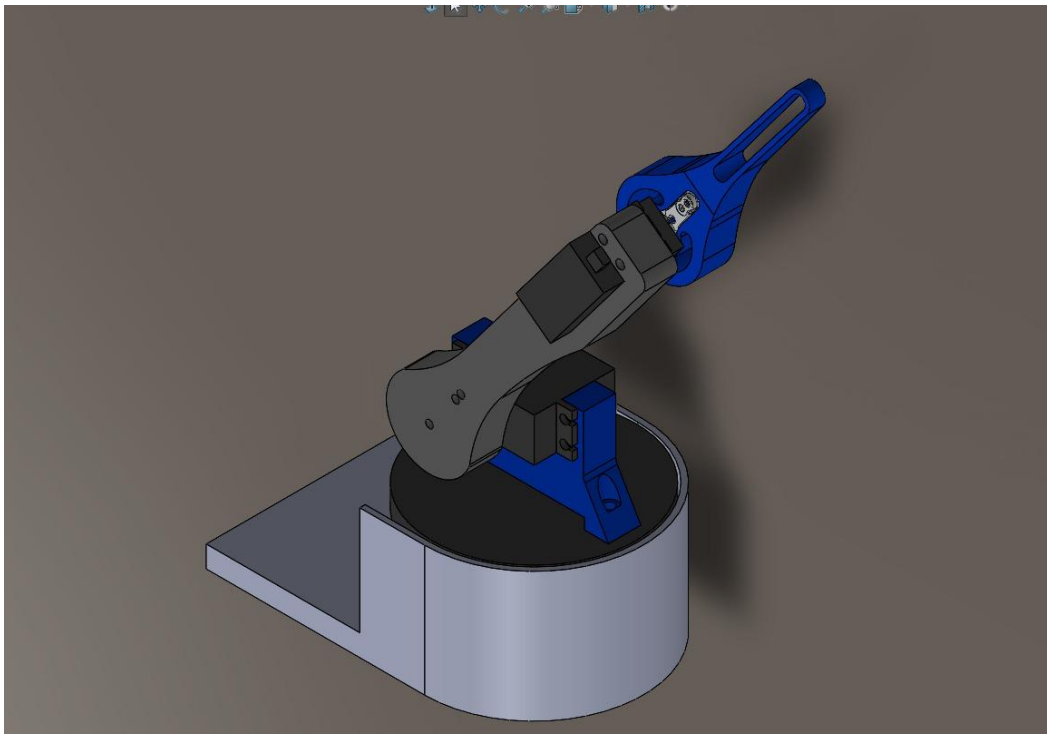
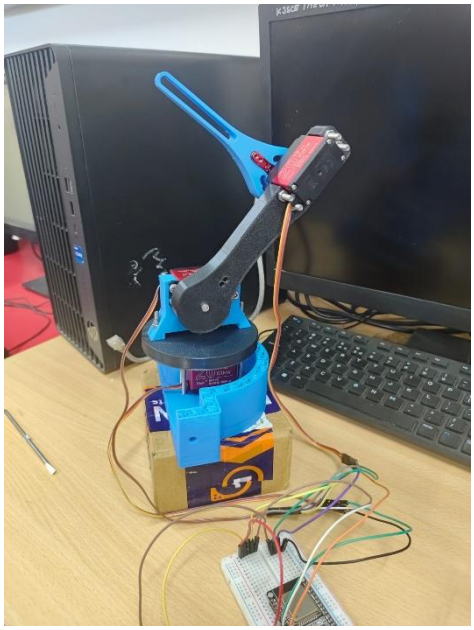
3-DOF Robotic Arm Prototype

PC-Based Forward/Inverse Kinematics and ESP32 Servo Control





Project Overview

This project develops a three-degree-of-freedom (3-DOF) robotic arm as a functional prototype for injection moulding automation. The arm uses three rotary joints to position its tip in three-dimensional space. The objective is to demonstrate robotic motion, joint-level actuation and PC-assisted kinematic control on a compact, low-cost platform.

Hardware and Control Architecture

Manipulator	3-DOF rotary-joint robotic arm
Actuators	20 kg·cm advertised class; 6.5 kg·cm rating used for design verification (supply-voltage dependent)
Microcontroller	ESP32 development board
Communication	UART serial interface between PC and ESP32
PC application	Forward kinematics (FK), inverse kinematics (IK) and joint-angle command generation

Working Principle

The operator specifies either a target end-effector position or desired joint angles in the PC application. For Cartesian targets, the application computes the required joint angles using inverse kinematics; forward kinematics is used to calculate or verify the resulting tip position. The PC sends the joint-angle commands to the ESP32 over UART. The ESP32 interprets the commands and generates the servo-control signals to move each joint to its commanded position.

Intended Application

The prototype is a learning and validation platform for future injection moulding robotic arms, particularly for repetitive pick-and-place tasks such as removing moulded parts and transferring them to a collection area. The present system demonstrates positioning and control principles; industrial deployment would require additional work on gripping, payload verification, repeatability, sensing and machine safety.

Preliminary Static Torque Verification

Assumptions: horizontal arm (worst-case gravity loading); no gripper or payload; shoulder link 100 mm/30 g; elbow link 105 mm/26 g; circular base link 30 g. Each servo is provisionally assumed to weigh 72 g, pending confirmation against the exact servo datasheet. Link centres of mass are approximated at their midpoints; actual CAD centre-of-mass values should replace these estimates.

Torque is reported in kgf·cm (commonly labelled kg·cm on servo datasheets). Static gravity torque = Σ[m(kg) × horizontal lever arm(cm)], with mass in kg giving numerical torque in kgf·cm.

Elbow (J3): elbow link = 0.026 × 5.25 = 0.137 kgf·cm. If a payload is later mounted, add payload mass (kg) × 10.5 cm. This estimate assumes the elbow servo rotates only the distal link.

Shoulder (J2): shoulder link = 0.030 × 5.0 = 0.150 kgf·cm; elbow link = 0.026 × (10.0 + 5.25) = 0.397 kgf·cm; elbow-mounted servo = 0.072 × 10.0 = 0.720 kgf·cm. Total ≈ 1.267 kgf·cm (≈ 0.124 N·m). The distal servo is assumed located at the elbow axis. Additional moving brackets, horns, wiring and payload are excluded.

Comparison with 6.5 kgf·cm rated servo torque: elbow utilisation ≈ 2.1% (static margin ≈ 47.6×); shoulder utilisation ≈ 19.5% (static margin ≈ 5.1×). These are ideal static estimates, not verified continuous-duty or dynamic margins. Stall torque, if that is what 6.5 kgf·cm represents, must not be treated as permissible continuous operating torque.

Base yaw (J1): gravity produces approximately zero torque about a vertical yaw axis in the ideal level configuration. Sizing requires rotational inertia, angular acceleration, friction and any eccentric loading.

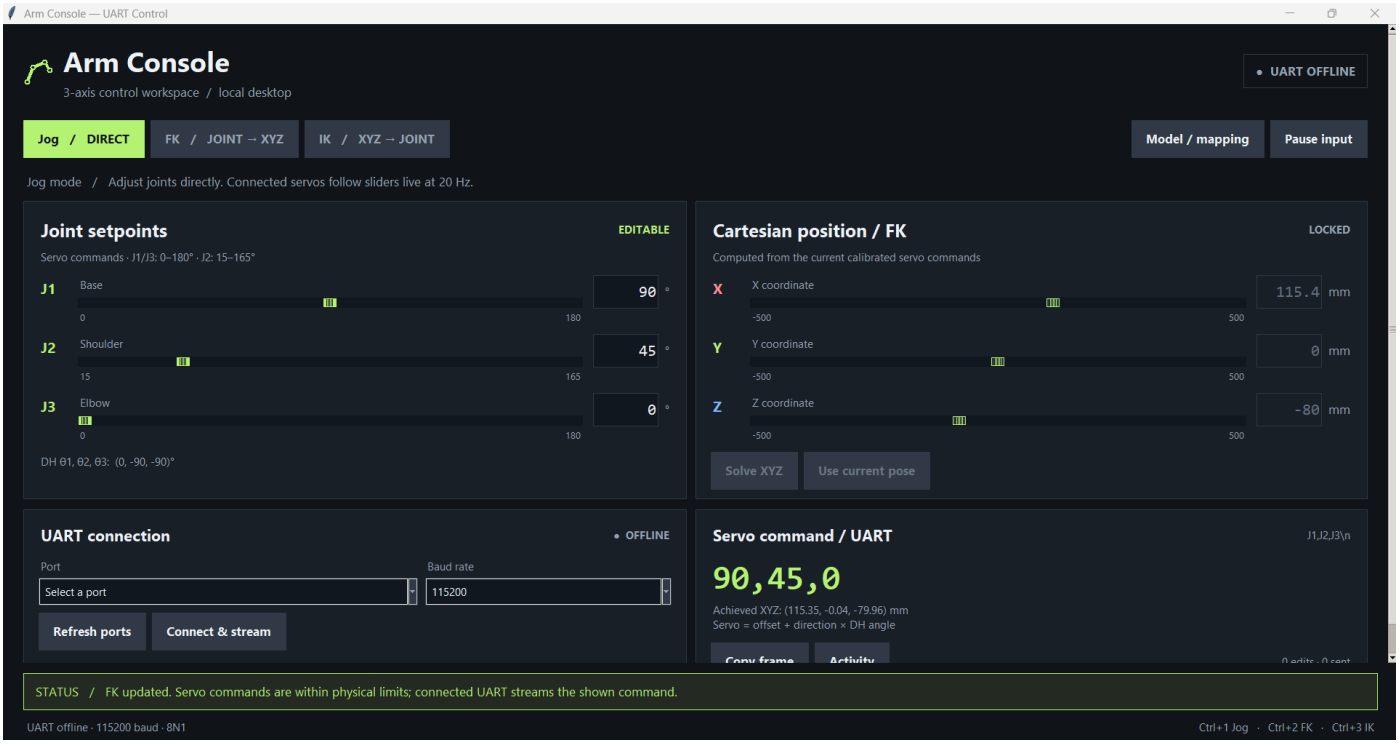
Verification status: provisionally adequate for unloaded static positioning at the shoulder and elbow, subject to confirming the exact servo model, operating voltage, torque rating type and 72 g mass, and validating CAD mass properties, acceleration loads and thermal duty cycle. A 2× design factor on the estimated shoulder gravity torque gives ≈ 2.53 kgf·cm, below 6.5 kgf·cm, but this does not establish continuous-duty suitability.

BOM		
Component	**Quantity**	**Cost**
Servo 20kgcm	3	₹ 3,600.00
esp32	1	₹ 400.00
SMPS 150W	1	₹ 1,500.00
M4 bolts	10	₹ 100.00
M4 nuts	10	₹ 40.00
M3 bolts	3	₹ 12.00
Buck converter	1	₹ 800.00
Jumper	1	₹ 40.00
	Total:	**₹ 6,492.00**

Working video:

https://drive.google.com/drive/folders/1AqqTZfMsxSqXxLJS2wNhXL85hsZ7pLjU?usp=sharing

GUI:



Team Members

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